

Lecture 19 Revision

L19.pdf

PDF

Please generate a lecture scribe that serves as a high-density revision tool. As you write, I want you to explicitly reason through the content step-by-step. This means you must include the reasoning directly in the scribe, explaining every definition and assumption clearly. If the material contains any proofs, derivations, or example solutions, walk through them exactly as they were presented in the context. Please make the logical flow highly explicit so that I can easily follow the sequence of thoughts during my final revision. It is critical that you do not introduce any outside material, new examples, or alternative explanations; limit your reasoning and content solely to what is present in the provided text. My goal is a concise, logical map of the actual lecture that I can refer to just before my exam.

Here is your **high-density lecture scribe (Lecture 19)**, written exactly as a **step-by-step logical reconstruction of the lecture**, strictly following the provided slides and preserving the reasoning flow.



Lecture 19: N-Variate Joint PDF / PMF / CDF

(Structured Revision Scribe)

Source:

L19

1. Joint Distribution of N Random Variables

Step 1: Representation

- The lecture begins by defining **multiple random variables together**.
- If we have $X_1, X_2, \dots, X_N$, we represent them as a **vector**:

$$X = [X_1, X_2, \dots, X_N]^T$$

Reasoning:

- Instead of studying variables separately, we group them to analyze **combined probabilistic behavior**.
- This is necessary because variables may be **dependent**.

Key Conclusion:

- The **joint distribution** describes probability behavior of all variables together.
- It can be represented using:
 - Joint PMF (discrete)
 - Joint PDF (continuous)

- Joint CDF
(mentioned explicitly on page 17)

2. Joint PMF, CDF, PDF (Definitions + Logic)

2.1 Joint PMF (Discrete Case)

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, \dots, X_N = x_N)$$

Reasoning:

- This directly extends single-variable PMF.
- Instead of one event, we consider **simultaneous occurrence of multiple events**.

2.2 Joint CDF

$$F_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

Reasoning:

- CDF accumulates probability.
- For multiple variables → accumulation happens over a **multi-dimensional region**.

2.3 Joint PDF

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = \frac{\partial^N F}{\partial x_1 \cdots \partial x_N}$$

Reasoning:

- PDF is obtained by differentiating the joint CDF.
- This mirrors the 1D case but extended to **N dimensions**.

3. Independence of Random Variables

Definition:

Variables are independent if:

$$f_X(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

Reasoning:

- If knowing one variable does not affect others $\rightarrow$ probabilities factorize.
- This is used later as a **test for independence**.

4. Example 1: Joint Gaussian Distribution

(From pages 4–16)

Step 1: Given Data

- $\mu_X = 2, \sigma_X^2 = 1$
- $\mu_Y = -3, \sigma_Y^2 = 4$
- $\text{cov}(X, Y) = -1$

Step 2: Need Correlation Coefficient

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Reasoning:

- Joint Gaussian PDF requires **correlation coefficient**, not covariance.

Compute standard deviations:

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Substitute:

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

Step 3: Substitute into Joint Gaussian Formula

Using general formula (page 9), substitute:

- $\mu_X = 2, \mu_Y = -3$
- $\sigma_X = 1, \sigma_Y = 2$
- $\rho = -0.5$

➡ This yields the **final joint PDF** (page 10).

Step 4: Marginal PDFs

Definition:

$$f_X(x) = \int f_{XY}(x, y) dy, \quad f_Y(y) = \int f_{XY}(x, y) dx$$

Key Property:

- Since variables are jointly Gaussian:

$$X \sim N(\mu_X, \sigma_X^2), \quad Y \sim N(\mu_Y, \sigma_Y^2)$$

Reasoning:

- Marginals of joint Gaussian remain Gaussian.

Step 5: Probability $P(X < 0)$

Step 5.1: Convert to CDF

$$P(X < 0) = F_X(0)$$

Step 5.2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$P(X < 0) = P\left(Z < \frac{0 - 2}{1}\right) = P(Z < -2)$$

Step 5.3: Use symmetry

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$

So:

$$P(Z < -2) = P(Z > 2)$$

Step 5.4: Use Q-function

$$Q(x) = P(Z > x)$$

Final Answer:

$$P(X < 0) = Q(2)$$

5. Application: Large MIMO System

(From pages 18–21)

Setup:

- M transmit antennas, N receive antennas.

Reasoning:

- Each channel path h_{ij} is modeled as a **random variable**.
- Entire system is represented as matrix:

$$H = [h_{ij}]$$

Insight:

- This motivates need for **joint distributions of many variables**.

6. Example 2: Uniform 3D Vector

Step 1: Problem Setup

- $X = (X_1, X_2, X_3)$
- Region: cube

$$|x_1| \leq 1, |x_2| \leq 1, |x_3| \leq 1$$

- PDF:

$$f = C \text{ inside cube, } 0 \text{ otherwise}$$

Step 2: Find Constant C

Condition:

$$\iiint f dx_1 dx_2 dx_3 = 1$$

Substitute:

$$C \cdot (2)(2)(2) = 1$$

$$C = \frac{1}{8}$$

Step 3: Marginal f_{X_1, X_2}

$$\begin{aligned} f_{X_1, X_2}(x_1, x_2) &= \int_{-1}^1 \frac{1}{8} dx_3 \\ &= \frac{1}{8}(2) = \frac{1}{4} \end{aligned}$$

Step 4: Marginal $f_{X_1}(x_1)$

$$\begin{aligned} f_{X_1}(x_1) &= \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3 \\ &= \frac{1}{8}(2)(2) = \frac{1}{2} \end{aligned}$$

Step 5: Conditional PDF

Using definition:

$$f_{X_1|X_2, X_3} = \frac{f_{X_1, X_2, X_3}}{f_{X_2, X_3}}$$

From previous results:

$$= \frac{1/8}{1/4} = \frac{1}{2}$$

Step 6: Independence Check

Condition:

$$f_{joint} = f_{X_1} f_{X_2} f_{X_3}$$

Compute RHS:

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Compare:

- LHS = $1/8$
- RHS = $1/8$

Conclusion:

✓ Variables are **independent**

7. Example 3: Trivariate Exponential

Step 1: Given PDF

$$f(x, y, z) = Ke^{-(ax+by+cz)}, \quad x, y, z > 0$$

Step 2: Find K

Condition:

$$\iiint f = 1$$

Factorization:

$$K \left(\int_0^\infty e^{-ax} dx \right) \left(\int_0^\infty e^{-by} dy \right) \left(\int_0^\infty e^{-cz} dz \right) = 1$$

Each integral:

$$= \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$$

Thus:

$$K \cdot \frac{1}{abc} = 1 \Rightarrow K = abc$$

Step 3: Marginal $f_{XY}(x, y)$

$$\begin{aligned} f_{XY}(x, y) &= \int_0^\infty abce^{-(ax+by+cz)} dz \\ &= abe^{-(ax+by)} \end{aligned}$$

Step 4: Marginal $f_X(x)$

$$\begin{aligned} f_X(x) &= \int_0^\infty abe^{-(ax+by)} dy \\ &= ae^{-ax} \end{aligned}$$

Step 5: Independence Check

LHS:

$$f_{XYZ} = abce^{-(ax+by+cz)}$$

RHS:

$$\begin{aligned} f_X f_Y f_Z &= (ae^{-ax})(be^{-by})(ce^{-cz}) \\ &= abce^{-(ax+by+cz)} \end{aligned}$$

Conclusion:

✓ LHS = RHS → variables are **independent**



Final Logical Map (Ultra-Compressed)

1. Define joint variables → vector form
2. Define PMF, CDF, PDF (multi-dimensional extension)
3. Independence → factorization condition
4. Gaussian example → compute p → substitute → marginals → standardization → Q-function
5. Uniform cube → normalization → marginals → conditional → independence
6. Exponential → factorization → marginals → independence

If you want, I can convert this into a **1-page formula sheet** or **exam cheat flowchart** for even faster revision.

Sources

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PDF

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Here is your **high-density, step-by-step reasoning lecture scribe** for Lecture 19, strictly reconstructed from the slides with **explicit logical flow, no added material**, and aligned exactly with how the lecture progresses.



Lecture 19: N-Variate Joint PDF / PMF / CDF

Source:

L19

1. Joint Distribution of N Random Variables

Step 1: Why we need a joint representation

- The lecture introduces multiple random variables $X_1, X_2, \dots, X_N$.
- Instead of treating them separately, we group them into a **vector form**:

$$X = [X_1, X_2, \dots, X_N]^T$$

Reasoning:

- This grouping is necessary because we want to study their **combined behavior**, not individual behavior.
- A single-variable distribution is insufficient when variables interact.

Step 2: What joint distribution represents

- The joint distribution:
 - Describes probability behavior of **all variables together**
 - Can be expressed using:
 - Joint PMF
 - Joint PDF
 - Joint CDF
- (as stated on page 17)

2. Joint PMF, CDF, PDF (Definition + Logical Meaning)

2.1 Joint PMF (Discrete Case)

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, X_2 = x_2, \dots, X_N = x_N)$$

Reasoning:

- This is a **direct extension of single-variable PMF**.
- Instead of one condition, we impose **multiple simultaneous conditions**.

2.2 Joint CDF

$$F_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

Reasoning:

- CDF accumulates probability.
- Here, accumulation is over a **multi-dimensional region**.

2.3 Joint PDF

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = \frac{\partial^N F}{\partial x_1 \partial x_2 \dots \partial x_N}$$

Reasoning:

- PDF is obtained by differentiating the joint CDF.
- This follows the same logic as 1D, extended to N variables.

3. Independence of Random Variables

Definition (page 25):

$$f_X(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

Reasoning:

- Independence means:
 - Knowing one variable gives **no information about others**
- Therefore, the joint distribution must **factor into marginals**

4. Example 1: Joint Gaussian Variables

(From pages 4-16)

Step 1: Given Data

- $\mu_X = 2, \sigma_X^2 = 1$
- $\mu_Y = -3, \sigma_Y^2 = 4$
- $\text{cov}(X, Y) = -1$

Goal:

- Joint PDF
- Marginals
- $P(X < 0)$

Step 2: Why correlation coefficient is needed

Before writing joint PDF, the slide states:

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Reasoning:

- The joint Gaussian formula uses **correlation coefficient**, not covariance.
- So conversion is necessary.

Compute standard deviations:

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Substitute:

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

Step 3: Construct Joint PDF

- Use the **general joint Gaussian formula** (shown on page 9).
- Substitute:
 - $\mu_X = 2$
 - $\mu_Y = -3$
 - $\sigma_X = 1$
 - $\sigma_Y = 2$
 - $\rho = -0.5$

➡ This produces the **final joint PDF** (page 10).

Step 4: Marginal PDFs

Definition:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

Key Property (explicitly stated):

- Since X, Y are jointly Gaussian:

$$X \sim N(\mu_X, \sigma_X^2), \quad Y \sim N(\mu_Y, \sigma_Y^2)$$

Reasoning:

- The lecture directly states marginals remain Gaussian.
- So integration is conceptually required, but result is known.

Step 5: Compute $P(X < 0)$

Step 5.1: Interpret as CDF

$$P(X < 0) = F_X(0)$$

Reasoning:

- By definition, CDF:

$$F_X(x) = P(X \leq x)$$

Step 5.2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

Substitute:

$$P(X < 0) = P\left(Z < \frac{0 - 2}{1}\right) = P(Z < -2)$$

Step 5.3: Use symmetry of normal CDF

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$

Thus:

$$P(Z < -2) = P(Z > 2)$$

Step 5.4: Use Q-function

Definition:

$$Q(x) = P(Z > x)$$

Final Answer:

$$P(X < 0) = Q(2)$$

5. Application: Large MIMO System

(From pages 18–21)

Step 1: System Setup

- M transmit antennas, N receive antennas.

Step 2: Modeling

- Each channel coefficient h_{ij} is a **random variable**.

Step 3: Matrix Representation

$$H = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \cdots & h_{MN} \end{bmatrix}$$

Reasoning:

- The entire system consists of **many random variables together**.
- This motivates studying **joint distributions of N variables**.

6. Example 2: Uniformly Distributed 3D Vector

Step 1: Setup

- $X = (X_1, X_2, X_3)$
- Region:

$$|x_1| \leq 1, |x_2| \leq 1, |x_3| \leq 1$$

- PDF:

$$f = C \text{ inside cube, } 0 \text{ outside}$$

Step 2: Find Constant C

Condition:

$$\iiint f dx_1 dx_2 dx_3 = 1$$

Substitute:

$$\begin{aligned} \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C dx_1 dx_2 dx_3 &= 1 \\ C(2)(2)(2) &= 1 \\ C &= \frac{1}{8} \end{aligned}$$

Step 3: Marginal f_{X_1, X_2}

$$\begin{aligned} f_{X_1, X_2}(x_1, x_2) &= \int_{-1}^1 \frac{1}{8} dx_3 \\ &= \frac{1}{8}(2) = \frac{1}{4} \end{aligned}$$

Step 4: Marginal $f_{X_1}(x_1)$

$$\begin{aligned} f_{X_1}(x_1) &= \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3 \\ &= \frac{1}{8}(2)(2) = \frac{1}{2} \end{aligned}$$

Step 5: Conditional PDF (Part d)

$$f_{X_1|X_2, X_3} = \frac{f_{X_1, X_2, X_3}}{f_{X_2, X_3}}$$

Using known values:

$$= \frac{1/8}{1/4} = \frac{1}{2}$$

Step 6: Conditional PDF (Part e)

$$f_{X_1, X_2|X_3} = \frac{f_{X_1, X_2, X_3}}{f_{X_3}}$$

Using symmetry result:

$$= \frac{1/8}{1/2} = \frac{1}{4}$$

Step 7: Independence Check

Condition:

$$f_{X_1, X_2, X_3} = f_{X_1} f_{X_2} f_{X_3}$$

LHS:

$$= \frac{1}{8}$$

RHS:

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion:

- LHS = RHS
✓ Variables are **independent**

7. Example 3: Trivariate Random Variables

Step 1: Given

$$f_{XYZ}(x, y, z) = K e^{-(ax+by+cz)}, \quad x, y, z > 0$$

Step 2: Find K

Condition:

$$\iiint f = 1$$

Factorization (key reasoning step):

- Exponential separates:

$$e^{-(ax+by+cz)} = e^{-ax} e^{-by} e^{-cz}$$

Thus:

$$K \left(\int_0^\infty e^{-ax} dx \right) \left(\int_0^\infty e^{-by} dy \right) \left(\int_0^\infty e^{-cz} dz \right) = 1$$

Each integral:

$$= \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$$

Final:

$$K \cdot \frac{1}{abc} = 1 \Rightarrow K = abc$$

Step 3: Marginal $f_{XY}(x, y)$

$$\begin{aligned} f_{XY}(x, y) &= \int_0^\infty abce^{-(ax+by+cz)} dz \\ &= abe^{-(ax+by)} \end{aligned}$$

Step 4: Marginal $f_X(x)$

$$\begin{aligned} f_X(x) &= \int_0^\infty abe^{-(ax+by)} dy \\ &= ae^{-ax} \end{aligned}$$

Step 5: Independence Check

LHS:

$$f_{XYZ} = abce^{-(ax+by+cz)}$$

RHS:

$$\begin{aligned} f_X f_Y f_Z &= (ae^{-ax})(be^{-by})(ce^{-cz}) \\ &= abce^{-(ax+by+cz)} \end{aligned}$$

Conclusion:

✓ LHS = RHS $\rightarrow$ variables are **independent**

FINAL REVISION FLOW (1-Minute Mental Map)

1. Vector representation → joint behavior
2. PMF / CDF / PDF → multi-variable extension
3. Independence → product of marginals
4. Gaussian:
 - Find ρ → substitute → marginals → standardize → Q
5. Uniform cube:
 - Normalize → integrate → conditionals → independence
6. Exponential:
 - Factorize → normalize → marginals → independence

If you want, I can compress this into a **single-page handwritten-style cheat sheet** or **flowchart for last 5-minute revision**.

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Lecture 19: N-Variate Joint PDF / PMF / CDF

Source:

L19

1. Representation of Multiple Random Variables

Step 1: Definition (page 17)

For $X_1, X_2, \dots, X_N$, define:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Reasoning:

- The lecture explicitly introduces a **vector form** to group variables.
- This is required because:
 - We are no longer analyzing variables individually.
 - We want to describe their **combined probabilistic behavior**.

Step 2: Meaning of Joint Distribution

- The joint distribution:
 - Describes **probability behavior of all variables together**
 - Can be represented as:
 - Joint PMF
 - Joint PDF
 - Joint CDF

→ This establishes the **core goal of the lecture**.

2. Joint Distribution Functions

2.1 Joint PMF (Discrete) — (page 22)

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, \dots, X_N = x_N)$$

Reasoning:

- Direct extension of single-variable PMF.
- Instead of one event, we consider **simultaneous events**.

2.2 Joint CDF — (page 23)

$$F_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

Reasoning:

- CDF accumulates probability.
- Now accumulation happens over an **N-dimensional region**.

2.3 Joint PDF — (page 24)

$$f_{X_1, \dots, X_N} = \frac{\partial^N F}{\partial x_1 \cdots \partial x_N}$$

Reasoning:

- Same logic as 1D:
 - PDF is derivative of CDF
- Extended to multiple variables → take **partial derivatives**

3. Independence of Random Variables (page 25)

$$f_X(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

Reasoning:

- Independence means:
 - Joint behavior = product of individual behaviors
- Therefore:
 - If factorization holds → variables are independent

4. Example 1: Joint Gaussian Variables

Step 1: Given (page 4)

- $\mu_X = 2, \sigma_X^2 = 1$
- $\mu_Y = -3, \sigma_Y^2 = 4$
- $\text{cov}(X, Y) = -1$

Step 2: Why compute correlation coefficient (page 7)

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Reasoning:

- The **joint Gaussian formula** uses ρ , not covariance.
- Therefore, conversion is required before substitution.

Step 3: Compute standard deviations (page 8)

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Step 4: Substitute into correlation formula

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

Step 5: Construct Joint PDF (pages 9–10)

- Use **general joint Gaussian formula**.
- Substitute:
 - $\mu_X = 2$
 - $\mu_Y = -3$
 - $\sigma_X = 1$
 - $\sigma_Y = 2$
 - $\rho = -0.5$

→ This yields the **final joint PDF** shown on page 10.

Step 6: Marginal PDFs (pages 11–12)

Definition:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

Key Property (explicitly stated):

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

Reasoning:

- The lecture directly states:
 - Marginals of jointly Gaussian variables are **univariate Gaussian**

- Hence:
 - Integration step is conceptually required, but result follows directly.

Step 7: Compute $P(X < 0)$ (pages 13–16)

Step 7.1: Express as CDF

$$P(X < 0) = F_X(0)$$

Reason:

- By definition:

$$F_X(x) = P(X \leq x)$$

Step 7.2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$P(X < 0) = P\left(Z < \frac{0 - 2}{1}\right) = P(Z < -2)$$

Step 7.3: Use symmetry of normal distribution

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$

Thus:

$$P(Z < -2) = P(Z > 2)$$

Step 7.4: Use Q-function

$$Q(x) = P(Z > x)$$

Final Answer:

$$P(X < 0) = Q(2)$$

5. Application: Large MIMO System (pages 18–21)

Step 1: Setup

- M transmit antennas

- N receive antennas

Step 2: Modeling assumption

- Each channel path h_{ij} is a **random variable**

Step 3: Matrix representation

$$H = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \cdots & h_{MN} \end{bmatrix}$$

Reasoning:

- The system contains **many random variables simultaneously**
- Therefore:
 - Joint distribution framework is necessary

6. Example 2: Uniform 3D Vector (pages 26–39)

Step 1: Given

$$|x_1| \leq 1, |x_2| \leq 1, |x_3| \leq 1$$

$$f = C \text{ inside cube, } 0 \text{ outside}$$

Step 2: Find C (page 28)

Condition:

$$\iiint f = 1$$

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C dx_1 dx_2 dx_3 = 1$$

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

Step 3: Marginal f_{X_1, X_2} (page 29)

$$\begin{aligned} f_{X_1, X_2}(x_1, x_2) &= \int_{-1}^1 \frac{1}{8} dx_3 \\ &= \frac{1}{8}(2) = \frac{1}{4} \end{aligned}$$

Step 4: Marginal f_{X_1} (page 31)

$$\begin{aligned} f_{X_1}(x_1) &= \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3 \\ &= \frac{1}{8}(2)(2) = \frac{1}{2} \end{aligned}$$

Step 5: Conditional PDF (d) (page 33)

$$\begin{aligned} f_{X_1|X_2, X_3} &= \frac{f_{X_1, X_2, X_3}}{f_{X_2, X_3}} \\ &= \frac{1/8}{1/4} = \frac{1}{2} \end{aligned}$$

Step 6: Conditional PDF (e) (page 35)

$$f_{X_1, X_2|X_3} = \frac{1/8}{1/2} = \frac{1}{4}$$

Step 7: Independence Check (pages 37–39)

Condition:

$$f_{X_1, X_2, X_3} = f_{X_1} f_{X_2} f_{X_3}$$

LHS:

$$= \frac{1}{8}$$

RHS:

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion:

✓ LHS = RHS → variables are **independent**

7. Example 3: Trivariate Exponential (pages 40–50)

Step 1: Given

$$f_{XYZ}(x, y, z) = K e^{-(ax+by+cz)}, \quad x, y, z > 0$$

Step 2: Find K (pages 41–42)

Condition:

$$\iiint f = 1$$

Key reasoning:

- Exponential separates:

$$e^{-(ax+by+cz)} = e^{-ax} e^{-by} e^{-cz}$$

Thus integral becomes product:

$$K \left(\int_0^\infty e^{-ax} dx \right) \left(\int_0^\infty e^{-by} dy \right) \left(\int_0^\infty e^{-cz} dz \right) = 1$$

Evaluate integrals:

$$= \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$$

Final:

$$K \cdot \frac{1}{abc} = 1 \Rightarrow K = abc$$

Step 3: Marginal $f_{XY}(x, y)$ (page 44)

$$\begin{aligned} f_{XY}(x, y) &= \int_0^\infty abce^{-(ax+by+cz)} dz \\ &= abe^{-(ax+by)} \end{aligned}$$

Step 4: Marginal $f_X(x)$ (page 46)

$$\begin{aligned} f_X(x) &= \int_0^\infty abe^{-(ax+by)} dy \\ &= ae^{-ax} \end{aligned}$$

Step 5: Independence Check (pages 47-50)

LHS:

$$f_{XYZ} = abce^{-(ax+by+cz)}$$

RHS:

$$\begin{aligned} f_X f_Y f_Z &= (ae^{-ax})(be^{-by})(ce^{-cz}) \\ &= abce^{-(ax+by+cz)} \end{aligned}$$

Conclusion:

✓ LHS = RHS → variables are **independent**

FINAL LOGICAL FLOW (Ultra-Condensed)

1. Represent variables → vector form
2. Define joint PMF / CDF / PDF
3. Independence → factorization rule
4. Gaussian:
 - cov → ρ → substitute → marginals → standardize → Q
5. Uniform cube:
 - normalize → integrate → conditional → independence
6. Exponential:
 - factorize → normalize → marginals → independence

If you want, I can turn this into a **one-page formula sheet + derivation triggers (what to recall in exam)** or a **visual flowchart**.

Sources

L19.pdf

PDF

Please generate a lecture scribe that serves as a high-density revision tool. As you write, I want you to explicitly reason through the content step-by-step. This means you must include the reasoning directly in the scribe, explaining every definition and assumption clearly. If the material contains any proofs, derivations, or example solutions, walk through them exactly as they were presented in the context. Please make the logical flow highly explicit so that I can easily follow the sequence of thoughts during my final revision. It is critical that you do not introduce any outside material, new examples, or alternative explanations; limit your reasoning and content solely to what is present in the provided text. My goal is a concise, logical map of the actual lecture that I can refer to just before my exam.

Here is your **final, exam-oriented lecture scribe** built strictly from the uploaded lecture slides . I have preserved the **exact reasoning flow used in the lecture**, making every step explicit and sequential, with no external additions.^{L19}

LECTURE 19 — N-VARIATE JOINT DISTRIBUTIONS

(High-Density Logical Revision Map)

1. WHY JOINT DISTRIBUTIONS? (Concept Foundation)

Step 1: Representation (page 17)

We define:

$$\mathbf{X} = [X_1, X_2, \dots, X_N]^T$$

Reasoning:

- Instead of treating variables separately, we group them.
- This is necessary because:
 - The lecture explicitly states:
 - 👉 "Joint distribution describes probability behavior of all variables together."
- Therefore:
 - We move from **single-variable** → **multi-variable framework**

Step 2: Forms of Representation

Joint behavior can be expressed as:

- Joint PMF (discrete)
- Joint PDF (continuous)

- Joint CDF

Reasoning:

- These are direct extensions of single-variable definitions.
- The only change: we now consider **simultaneous events**

2. CORE DEFINITIONS

2.1 Joint PMF (page 22)

$$P_{X_1, \dots, X_N}(x_1, \dots, x_N) = P(X_1 = x_1, \dots, X_N = x_N)$$

Reasoning:

- This represents probability of **all variables taking specific values at once**
- It is not separate probabilities—it is **one combined event**

2.2 Joint CDF (page 23)

$$F(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

Reasoning:

- This accumulates probability over an **N-dimensional region**
- Extension of 1D CDF:
 - Instead of a line → now a region

2.3 Joint PDF (page 24)

$$f = \frac{\partial^N F}{\partial x_1 \cdots \partial x_N}$$

Reasoning:

- Same logic as 1D:
 - PDF = derivative of CDF
- For N variables:
 - Take **N partial derivatives**

3. INDEPENDENCE CONDITION (page 25)

$$f(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

Reasoning:

- Lecture definition:
 - Variables are independent **if joint PDF equals product of marginals**
- This becomes the **test for independence**

4. EXAMPLE 1 — JOINT GAUSSIAN

Step 1: Given (page 4)

- $\mu_X = 2, \sigma_X^2 = 1$
- $\mu_Y = -3, \sigma_Y^2 = 4$
- $\text{cov}(X, Y) = -1$

Step 2: Why compute correlation first (page 7)

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Reasoning:

- The joint Gaussian formula uses **correlation coefficient**
- Therefore:
 - Covariance must be converted to correlation

Step 3: Compute standard deviations (page 8)

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Step 4: Substitute into correlation

$$\rho = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

Step 5: Construct Joint PDF (pages 9–10)

- Use general Gaussian formula (shown on page 5)

- Substitute:
 - $\mu_X = 2, \mu_Y = -3$
 - $\sigma_X = 1, \sigma_Y = 2$
 - $\rho = -0.5$

→ This directly gives the **final joint PDF (page 10)**

Step 6: Marginal PDFs (pages 11–12)

Definition:

$$f_X(x) = \int f_{XY}(x, y) dy, \quad f_Y(y) = \int f_{XY}(x, y) dx$$

Key Property (explicit in lecture):

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

Reasoning:

- Lecture states:
 - Marginals of jointly Gaussian variables are **Gaussian**
- Therefore:
 - Integration step is conceptually shown
 - Result is directly written

Step 7: Compute $P(X < 0)$ (pages 13–16)

Step 7.1: Express as CDF

$$P(X < 0) = F_X(0)$$

Reason:

- By definition:
 - $F_X(x) = P(X \leq x)$

Step 7.2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$P(X < 0) = P\left(Z < \frac{0 - 2}{1}\right) = P(Z < -2)$$

Step 7.3: Use symmetry (page 15)

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$

Thus:

$$P(Z < -2) = P(Z > 2)$$

Step 7.4: Q-function (page 16)

$$Q(x) = P(Z > x)$$

Final Answer:

$$P(X < 0) = Q(2)$$

5. APPLICATION — LARGE MIMO SYSTEM (pages 18–21)

Step 1: System

- M transmit antennas
- N receive antennas

Step 2: Modeling assumption

Each channel:

$$h_{ij} = \text{random variable}$$

Step 3: Matrix form

$$H = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \cdots & h_{MN} \end{bmatrix}$$

Reasoning:

- System contains **many random variables simultaneously**
- Therefore:
 - Requires joint distribution modeling

6. EXAMPLE 2 — UNIFORM 3D VECTOR

Step 1: Given (page 26)

$$|x_1| \leq 1, |x_2| \leq 1, |x_3| \leq 1$$

$$f = \begin{cases} C, & \text{inside cube} \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Find C (page 28)

Condition:

$$\iiint f = 1$$

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C = 1$$

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

Step 3: Marginal f_{X_1, X_2} (page 29)

$$\int_{-1}^1 \frac{1}{8} dx_3 = \frac{1}{4}$$

Step 4: Marginal f_{X_1} (page 31)

$$\int_{-1}^1 \int_{-1}^1 \frac{1}{8} = \frac{1}{2}$$

Step 5: Conditional PDF (page 33)

$$f_{X_1|X_2, X_3} = \frac{1/8}{1/4} = \frac{1}{2}$$

Step 6: Conditional PDF (page 35)

$$f_{X_1, X_2|X_3} = \frac{1/8}{1/2} = \frac{1}{4}$$

Step 7: Independence Check (pages 37–39)

Condition:

$$f_{X_1, X_2, X_3} = f_{X_1} f_{X_2} f_{X_3}$$

LHS:

$$= \frac{1}{8}$$

RHS:

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion:

✓ LHS = RHS → independent

7. EXAMPLE 3 — TRIVARIATE EXPONENTIAL

Step 1: Given (page 40)

$$f_{XYZ} = K e^{-(ax+by+cz)}, \quad x, y, z > 0$$

Step 2: Find K (pages 41–42)

Condition:

$$\iiint f = 1$$

Key reasoning:

- Exponential separates:

$$e^{-(ax+by+cz)} = e^{-ax} e^{-by} e^{-cz}$$

Integral becomes:

$$K \left(\frac{1}{a} \right) \left(\frac{1}{b} \right) \left(\frac{1}{c} \right) = 1$$

Final:

$$K = abc$$

Step 3: Marginal f_{XY} (page 44)

$$f_{XY} = abe^{-(ax+by)}$$

Step 4: Marginal f_X (page 46)

$$f_X = ae^{-ax}$$

Step 5: Independence Check (pages 47–50)

LHS:

$$abce^{-(ax+by+cz)}$$

RHS:

$$(ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Conclusion:

✓ LHS = RHS → **independent**

FINAL REVISION FLOW (MENTAL MAP)

1. Vector form → joint behavior
2. PMF / CDF / PDF definitions
3. Independence → product rule
4. Gaussian:
 - cov → ρ → substitute → marginals → standardize → Q
5. Uniform cube:
 - normalize → integrate → conditional → independence
6. Exponential:

- factorize → normalize → marginals → independence
-

If you want next: I can compress this into a **1-page cheat sheet or last-night revision triggers (what to recall in 30 sec per topic)**.

Sources